Lab 3 Solution

```r
library(tidyverse)

ncbikecrash <- read_csv("https://karamccor.github.io/b6002/labs/data/bikecrash.csv")
```

Exercise 1

(a)

```r
ncbikecrash |>
  filter(development == "Residential") |>
  filter(driver_age_group == "0-19")
```

There were 223 crashes in residential areas where the driver age group was under 0-19.

(b)

```r
ncbikecrash |>
  filter(development == "Residential") |>
  count(driver_age_group)
```

```
# A tibble: 9 x 2
  driver_age_group     n
  <chr>            <int>
1 0-19               223
2 20-24              364
3 25-29              290
4 30-39              508
5 40-49              441
6 50-59              372
```

```
7 60-69                269
8 70+                  243
9 <NA>                 457
```

Exercise 2

```
ncbikecrash |>
  count(driver_est_speed) |>
  arrange(n)
```

```
# A tibble: 17 x 2
   driver_est_speed     n
   <chr>            <int>
 1 81-85 mph            1
 2 >100 mph             2
 3 66-70 mph            4
 4 61-65 mph           10
 5 56-60 mph           29
 6 46-50 mph          142
 7 36-40 mph          261
 8 51-55 mph          338
 9 26-30 mph          367
10 21-25 mph          525
11 16-20 mph          544
12 11-15 mph          552
13 <NA>               580
14 41-45 mph          670
15 6-10 mph           840
16 31-35 mph          846
17 0-5 mph           1756
```

Exercise 3

```
ncbikecrash |>
  filter(county == "Durham" | county == "Orange" | county == "Wake") |>
  group_by(county) |>
  summarize(mean_driver_age = mean(driver_age, na.rm = T))
```

```
# A tibble: 3 x 2
  county mean_driver_age
  <chr>            <dbl>
1 Durham            40.4
2 Orange            38.3
3 Wake              38.2
```

Exercise 4

```
ncbikecrash |>
  filter(county == "Durham") |>
  group_by(crash_month) |>
  summarize(mean_crash_hour = mean(crash_hour)) |>
  arrange(mean_crash_hour)
```

```
# A tibble: 12 x 2
   crash_month mean_crash_hour
   <chr>                 <dbl>
 1 January                13.5
 2 August                 13.9
 3 September              14.0
 4 October                14.2
 5 November               14.4
 6 December               14.6
 7 May                    14.9
 8 March                  15.2
 9 July                   15.3
10 April                  15.3
11 June                   16.2
12 February               16.8
```

Exercise 5

```
ncbikecrash_time <- ncbikecrash |>
  mutate(time_crash = case_when(
    crash_hour < 7 ~ "Early Morning",
    crash_hour >=7 & crash_hour < 12 ~ "Morning",
    crash_hour >=12 & crash_hour < 18 ~ "Afternoon",
```

```r
    crash_hour >=18 ~ "Evening"
  ))

ncbikecrash_time |>
  count(time_crash)
```

```
# A tibble: 4 x 2
  time_crash        n
  <chr>         <int>
1 Afternoon      3412
2 Early Morning   425
3 Evening        2270
4 Morning        1360
```

Exercise 6

```r
ncbikecrash |>
  summarize(mean_driver_age = mean(driver_age, na.rm = T))
```

```
# A tibble: 1 x 1
  mean_driver_age
            <dbl>
1            39.0
```

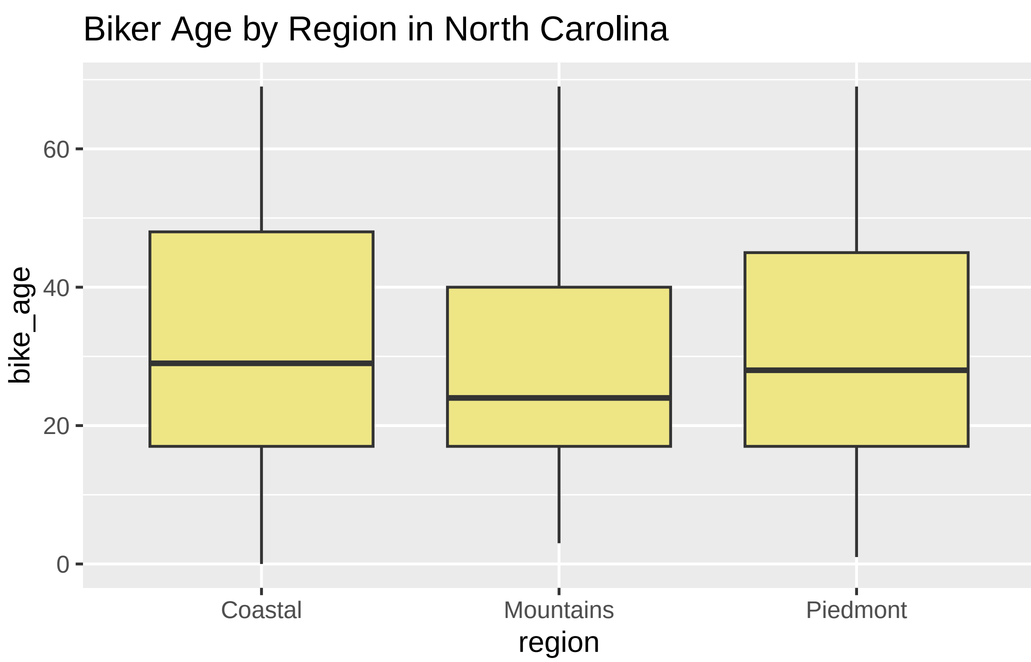
```r
ncbikecrash |>
  group_by(county) |>
  summarize(mean_driver_age = mean(driver_age, na.rm = T)) |>
  arrange(desc(mean_driver_age)) |>
  slice(1:5)
```

```
# A tibble: 5 x 2
  county   mean_driver_age
  <chr>              <dbl>
1 Camden              57.8
2 Stokes              55.2
3 Mitchell            54
4 Anson               49.6
5 Polk                47
```

Exercise 7

(a)

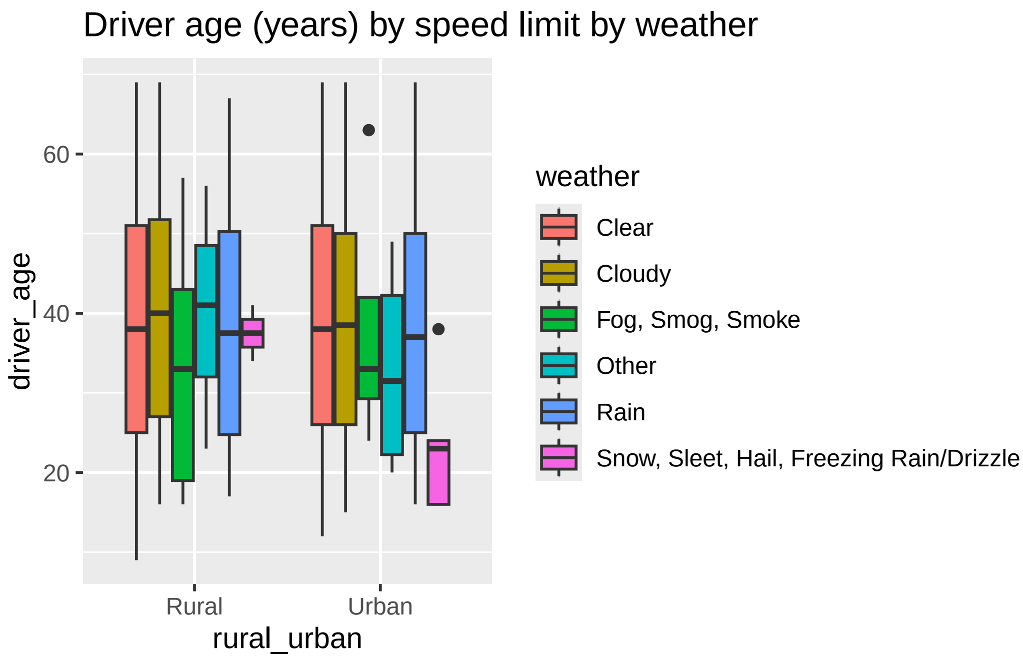
```
ggplot(ncbikecrash, mapping = aes(x = region, y = bike_age)) +
  geom_boxplot(fill = "khaki2") +
  labs(title = "Biker Age by Region in North Carolina",
       xlab = "Urban vs. Rural area",
       ylab = "Driver age (years)")
```



The biker ages seem to be relatively similar across groups.

(b)

```
ggplot(ncbikecrash, mapping = aes(x = rural_urban, y = driver_age, fill = weather)) +
  geom_boxplot() +
  labs(title = "Driver age (years) by speed limit by weather",
       xlab = "Rural vs. Urban area",
       ylab = "Driver Age (years)")
```



For Urban areas with `Snow, Sleet, Hail, Freezing Rain/Drizzle`, drivers involved in the crashes tended to be younger, with a median around 23 years old, compared to Rural areas where the same group was older (median age around 36 years.)